FLOCK ENERGY

Software Development Assignment Submission

BACKEND ENGINEERING

Backend API Service for Urja Meter Operations

CANDIDATE DETAILS

NAME
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ROLE
Software Development

GitHub Repository

https://github.com/Arnav-Nehra28/flock_energy_assignment_arnav

View on GitHub →

PROJECT OVERVIEW

Developed a production-ready Python **FastAPI** backend service that provides clean, structured, and typed REST API access to meter operations data. The service acts as an abstraction layer over the legacy Urja Meter Ops portal, handling automatic session renewal, CSRF mitigation, dynamic HMAC-SHA256 request signing for bulk data exports, normalization across legacy and v2 meter schemas, multi-attribute filtering, and 7-tier distribution network hierarchy reconstruction.

SUBMISSION CONTENTS

✓ **Source Code & API**	FastAPI service with modular routers (`meters`, `transformers`, `hierarchy`), Pydantic v2 schemas, and async portal client.
✓ **README.md**	Front-door documentation covering setup, architecture, teammate handover guide, error mapping policies (404/502/503), and trade-offs.
✓ **PROTOCOL.md**	Reverse-engineering documentation analyzing SvelteKit SSR, Better-Auth cookies, HMAC signing scheme, and schema formats.
✓ **openapi.json**	Complete OpenAPI 3.1.0 specification with typed response envelopes and declared error responses.
✓ **REFLECTION.md**	In-depth engineering reflections covering technical assumptions, obstacles, mistakes made, and production roadmap.
✓ **Automated Tests**	30 deterministic offline unit and API tests, isolated live integration tests (`pytest -m live`), and GitHub Actions CI workflow.

EXECUTION INSTRUCTIONS

Refer to `README.md` in the repository for detailed configuration and execution options. To run locally:

```
# 1. Setup environment and install dependencies python3 -m venv venv && source venv/bin/activate && pip install -r requirements.txt cp .env.example .env # 2. Run deterministic test suite (30 passed in ~2s) pytest -v # 3. Start local API server (Swagger UI at http://localhost:8000/docs) uvicorn app.main:app --host 0.0.0.0 --port 8000 --reload
```